

Premium calculation based on quantiles

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Quantile regression

Quantile regression (Koenker & Bassett, 1978) specifies a linear relationship between the quantiles of the dependent variable S_i and the covariates x . The relationship between the θ – quantile of S_i and the covariates x is given by:

$$Q_{S_i}(\theta|x_i) = x_i\beta_\theta$$

where $Q_{S_i}(\theta|x_i)$ refers to the conditional quantile of S_i for a probability θ conditional on the covariates x .

Under the distribution-free approach, to obtain the estimates of the parameters for a sample of size n , it is necessary to solve:

$$\min_{\beta_\theta} \frac{1}{n} \left\{ \theta \sum_{i:s_i \geq x_i\beta_\theta} |s_i - x_i\beta_\theta| + (1 - \theta) \sum_{i:s_i < x_i\beta_\theta} |s_i - x_i\beta_\theta| \right\}$$

The quantreg R package (Koenker, 2015) provides the estimation algorithms for quantile regression models.

A two-stage model based on quantiles

- The **Two-Stage Quantile Regression model for premium calculation** was proposed by Heras et al. (2018).
 - We assume that we have a portfolio of I policyholders ($i=1, \dots, I$), N_i is the number of claims for policyholder i , S_i is the total claim amount for policyholder i , and $\mathbf{x}_i$ are the covariates.
 - We want to calculate the θ – quantile of S_i , $Q_{S_i}(\theta|\mathbf{x}_i)$ where $\theta = F_{S_i}(s_i|\mathbf{x}_i)$ is the cumulative probability that policyholder i is not going to exceed the total claim amount s_i .
 - We have that

$$\Pr(S_i \leq s_i | \mathbf{x}_i) = \Pr(N_i = 0 | \mathbf{x}_i) + \Pr(N_i > 0, S_i \leq s_i | \mathbf{x}_i)$$

$$\Pr(N_i > 0, S_i \leq s_i | \mathbf{x}_i) = \Pr(N_i > 0 | \mathbf{x}_i) \Pr(S_i \leq s_i | N_i > 0, \mathbf{x}_i)$$

$$\Pr(S_i \leq s_i | \mathbf{x}_i) = \Pr(N_i = 0 | \mathbf{x}_i) + \Pr(N_i > 0 | \mathbf{x}_i) \Pr(S_i \leq s_i | N_i > 0, \mathbf{x}_i)$$

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- Or, equivalently

$$F_{S_i}(s_i|x_i) = F_{N_i}(0|x_i) + (1 - F_{N_i}(0|x_i))F_{S_i|N_i>0}(s_i|x_i)$$

- Lets call $F_{N_i}(0|x_i) = p_i$ and $F_{S_i|N_i>0}(s_i|x_i) = \theta^*$. Then, we can rewrite the former expression as:

$$\theta = p_i + (1 - p_i) \cdot \theta^*$$

Or, equivalently

$$\theta^* = \frac{\theta - p_i}{1 - p_i}$$

- It is clear that $F_{S_i}(s_i|x_i) = \theta$ is equivalent to $F_{S_i|N_i>0}(s_i|x_i) = \theta^*$. In other words, the θ – quantile of S_i is the same as the θ^* – quantile of $S_i|N_i > 0$.

A two-stage model based on quantiles

- Therefore, the authors propose to estimate the model in **two-stages**:
 - **Stage 1**: A Logistic regression model is used to estimate that a given policy has no claims, p_i .
 - Note that the logistic regression model should take into account the risk exposition. Let us assume $y \sim B(1, \pi)$ where $E[y] = \pi$. Then, the Logistic regression is specified as $\log\left(\frac{\pi}{1-\pi}\right) = X\beta$.
 - We assume that the probability is proportional to the exposition, $\pi^* = t\pi$.
 - Then, $\log\left(\frac{\pi^*/t}{1-\pi^*/t}\right) = X\beta$, or equivalently, $\pi^* = t \cdot \frac{e^{X\beta}}{1+e^{X\beta}}$
 - This means that we should use a modified logit link in the model specification. This can be easily done by using PROC GENMOD in SAS, statements FWDLINK and INVLINK.

A two-stage model based on quantiles

- **Stage 2:** A Quantile regression model is applied for each policyholder type, according to his estimated probability of submitting at least one claim, calculated in Stage 1.

Note that:

- As we are conditioning on $N_i > 0$, it is necessary to remove zeroes from S_i .
- To get a multiplicative tariff (as we normally do with GLMs), the response variable of the quantile regression has to be the log of the aggregated costs, $\log(S_i)$ conditioned on having submitted at least one claim.
- The authors consider the level $\theta = 95\%$.

A two-stage model based on quantiles

- The authors apply this model for **premium calculation** based on the classical **Two-Part model approach**. Specifically, they compare **two principles** for premium calculation, one of them based on the proposed methodology:

- **Expected Value Premium Principle (EVPP):**

$$P_i = E(S_i) + \alpha_1 \cdot E(S_i) = (1 + \alpha_1) \cdot E(S_i)$$

- In that case, the expected value of the total claim cost for policyholder i is obtained by multiplying the probability of having at least one claim (logit model) by the expected total claim cost for policyholder i estimated by using a GLM (the authors used a Gamma model).
- This approach has been criticized because the premium loading does not depend on the risk.
- Alternative approaches are the Variance Premium Principle (VPP), where $P_i = E(S_i) + \alpha_1 \cdot \text{Var}(S_i)$, and the Standard Deviation Premium Principle (SDPP), where $P_i = E(S_i) + \alpha_1 \cdot \sqrt{\text{Var}(S_i)}$.

A two-stage model based on quantiles

- **Quantile Premium Principle (QPP):** By using the two stage model, the premium can be calculated as

$$P_i = E(S_i) + \alpha_2 \cdot [Q_{S_i}(\theta) - E(S_i)] = \alpha_2 \cdot Q_{S_i}(\theta) + (1 - \alpha_2) E(S_i)$$

- This principle has better properties compared to the EVPP, as the premium loading depends on the risk.
- Heras et al. (2018) compare both principles for $\theta = 95\%$, by using a well-known insurance data set.
- They authors choose $\alpha_1 = 11.96\%$ and $\alpha_2 = 3\%$, so that the total income for the insurance company would be similar for both premium principles.

References

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